Sahaj Bhadja

(213) 284-4651 | bhadja@usc.edu | sahaj-bhadja | sahaj-bhadja | sahaj-bhadja

Summary

Software engineer with 2+ years building healthcare workflows, enterprise REST APIs, ETL pipelines, and database-backed applications using Python, Java, React, SQL, AWS, Azure, automated testing, and Agile delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Architected multi-tenant **Azure SQL** and Blob Storage services for 500+ users, optimizing normalized schemas, indexes, and access patterns to reduce enterprise dashboard latency from 1.2 seconds to 350 milliseconds.
- Built **FastAPI ETL and integration workflows** processing 5,000+ monthly jobs with validation, reconciliation, secure error handling, and CI/CD-backed releases, achieving 99%+ reliability across production data operations.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered **Python REST APIs** and React workflows handling 10,000+ daily grading requests, integrating secure file ingestion, AWS RDS persistence, and automated processing while maintaining stable production performance.
- Designed concurrent batch processing for 1,000+ uploaded files using multithreading and fault-tolerant execution, reducing feedback turnaround by 60% while improving **debugging and defect resolution** during peak workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built reproducible **Python data pipelines** processing 100,000+ trajectory logs for autonomous-driving simulations, improving long-horizon planning by 18% through systematic testing, debugging, and reliable enterprise-scale model evaluation.
- Architected a spatial-temporal encoder combining HD-map geometry and historical trajectories through multi-head attention, improving closed-loop simulation stability by 15% across complex, high-volume, real-world data workflows.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform **microservices** with Jenkins CI/CD and 20+ PyTest and Mockito suites, automating production unit and integration testing across Windows, Linux, and macOS in Agile delivery cycles.
- Optimized **REST API and ETL integrations** using object-oriented patterns, AWS Glue, S3, and SQL, reducing response times by 30% and improving scalable enterprise data-pipeline efficiency by 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed a MERN resource-management platform with **REST APIs**, WebSockets, and eight modular services, supporting 2M+ synchronized records while translating secure business requirements into maintainable real-time enterprise workflows.
- Automated Docker-based builds, testing, and deployments through **Jenkins CI/CD**, reducing release time by 50% while improving automated code quality, documentation, and defect resolution across frequent platform releases.

Skills

Languages & Backend: Python, Java, JavaScript, TypeScript, SQL, C++, Bash, FastAPI, Flask, Spring Boot, Node.js, REST APIs, microservices, object-oriented programming; **Data & Frontend:** PostgreSQL, SQL Server, Azure SQL, MySQL, MongoDB, Firebase, AWS RDS, ETL, data integration, React, HTML5, CSS3, WebSockets; **Cloud & Delivery:** AWS, Azure, S3, AWS Glue, Docker, Kubernetes, Jenkins, Git, CI/CD, PyTest, Mockito, unit testing, integration testing, debugging, Agile/Scrum, technical documentation, HIPAA-aware development

Projects

HMIS Live | *TypeScript, MERN, ABDM APIs, Sentry, Twilio*
- Built a secure **TypeScript healthcare platform** integrating consent-based ABDM APIs, appointment workflows, Twilio alerts, and Sentry monitoring, reducing administrative workload by 30% across patient and clinician operations.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, REST APIs*
- Built an AI-assisted Java and TypeScript database workspace supporting 10+ systems, accelerating **SQL authoring**, schema exploration, validation, and enterprise relational data management through a unified cross-platform interface.

Linguistix | *Python, AWS, Docker, CI/CD*
- Developed and deployed a containerized **Python REST data-processing API** on AWS ECS and Fargate with CodePipeline automation, validation checks, and CI/CD releases, achieving 87% accuracy and production reliability.